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Nessus Essentials / OpenVAS Deployment, Metasploitable2 Scan & Vulnerability Severity Analysis

Phase 1: Installation & Setup Guide

Option A: Setting Up Nessus Essentials on Kali Linux

1. Download & Install Package

Download the Debian package (`.deb`) for Linux from the official Tenable website.

Install using:

Bash

```
sudo dpkg -i Nessus-*.deb
```

2. Start the Nessus Service

Bash

```
sudo systemctl start nessusd  
sudo systemctl enable nessusd
```

3. Initial Configuration

Open your browser and navigate to:

```
https://localhost:8834
```

Then:

- Select **Nessus Essentials**
- Register for a free activation key
- Enter the activation key

- Create an administrator username and password
 - Wait for plugin compilation to complete
-

Option B: Setting Up OpenVAS (Greenbone Community Edition)

1. Install OpenVAS

Bash

```
sudo apt update  
sudo apt install gvm -y
```

2. Perform Initial Setup

Bash

```
sudo gvm-setup
```

Note: Save the randomly generated administrator password displayed after setup.

3. Verify Installation

Bash

```
sudo gvm-check-setup  
sudo gvm-start
```

Open your browser:

```
https://127.0.0.1:9392
```

Phase 2: Scanning the Metasploitable2 Virtual Machine

1. Prerequisites

- Boot the **Metasploitable2** virtual machine.
- Ensure both Kali Linux and Metasploitable2 are on the same virtual network.

- Login using:

Username: msfadmin

Password: msfadmin

Find the target IP address:

Bash

ifconfig

Example:

10.0.2.8

2. Executing a Scan in Nessus

Create a New Scan

Navigate to:

Scans → New Scan

Choose:

Basic Network Scan

Configure:

Setting	Value
Name	Metasploitable2 Assessment
Target	10.0.2.8

Optional: Configure SSH Credentials

Navigate to:

Credentials → SSH

Authentication Method:

Password

Credentials:

Username: msfadmin

Password: msfadmin

Launch Scan

Click:

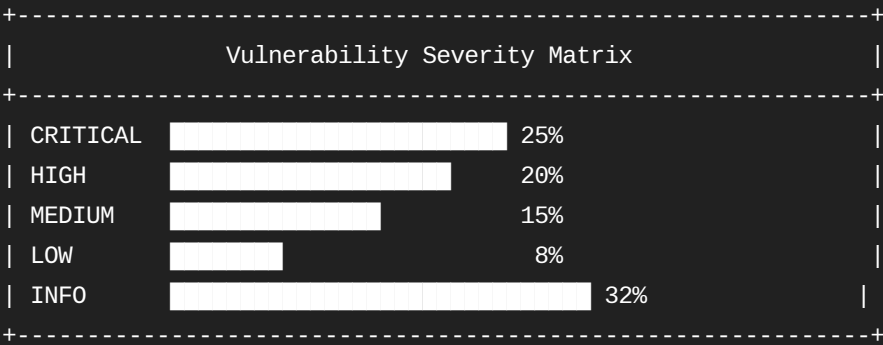
Save → Launch

Wait until the scan completes.

Phase 3: Vulnerability Report & Severity Analysis

After scanning, Nessus categorizes vulnerabilities according to **CVSS v3** severity ratings.

Vulnerability Severity Matrix



Vulnerability Findings

Severity	CVSS v3	Vulnerability	Service	Root Cause & Impact
Critical	9.0-10.0	vsftpd 2.3.4 Backdoor RCE	FTP (Port 21)	Backdoor allows unauthenticated

Severity	CVSS v3	Vulnerability	Service	Root Cause & Impact
				remote command execution as root.
Critical	9.0-10.0	Unauthenticated Bind Shell	Port 1524	Root shell accessible without authentication.
High	7.0-8.9	UnrealIRCd 3.2.8.1 Backdoor	IRC (Port 6667)	Malicious IRC commands execute arbitrary payloads.
High	7.0-8.9	Weak SSH Key Pair	SSH (Port 22)	Predictable OpenSSL random number generation enables key compromise.
Medium	4.0-6.9	Apache HTTP Server Multiple Vulnerabilities	Ports 80, 443	Outdated Apache version vulnerable to information disclosure and denial-of-service attacks.
Medium	4.0-6.9	Anonymous NFS Export	Port 2049	Root filesystem exported with read/write permissions to anonymous users.
Low	0.1-3.9	TCP Timestamp Information Disclosure	Network Stack	System uptime can be estimated remotely using TCP timestamps.

Severity Definitions

1. Critical Severity (CVSS 9.0-10.0)

Definition

Vulnerabilities that can be exploited remotely with little or no user interaction, allowing:

- Remote Code Execution (RCE)
- Full system compromise
- Administrative access

Recommended Remediation

- Apply security patches immediately.
 - Remove or replace vulnerable software.
 - Disable affected services if patching is not possible.
-

2. High Severity (CVSS 7.0-8.9)

Definition

Serious vulnerabilities enabling:

- Privilege escalation
- Authentication bypass
- Sensitive data theft
- Remote exploitation

Recommended Remediation

- Apply vendor patches as a high priority.
 - Restrict network access.
 - Enforce strong authentication.
-

3. Medium Severity (CVSS 4.0-6.9)

Definition

Vulnerabilities requiring specific conditions for exploitation that may lead to:

- Information disclosure
- Partial denial of service
- Limited unauthorized access

Recommended Remediation

- Update vulnerable software.
 - Restrict unnecessary services.
 - Secure exported resources such as NFS shares.
-

4. Low Severity (CVSS 0.1-3.9)

Definition

Minor weaknesses that reveal limited system information or require complex attack scenarios.

Recommended Remediation

- Harden system configurations.
 - Disable unnecessary network features.
 - Remove information leakage wherever practical.
-

Conclusion

The vulnerability assessment demonstrates that **Metasploitable2** intentionally contains numerous exploitable services for security testing and training. Critical vulnerabilities such as the **vsftpd 2.3.4 backdoor** and **Unauthenticated Bind Shell** allow complete system compromise without authentication, while High, Medium, and Low severity findings expose additional weaknesses affecting confidentiality, integrity, and availability.

Performing scans with **Nessus Essentials** or **OpenVAS (Greenbone Community Edition)** enables security professionals to identify vulnerable services, prioritize remediation using CVSS scores, and strengthen system security through timely patching, service hardening, and secure configuration practices.